

B6 Polynomials

Name: _____

Class: _____

Date: _____

Time: **377 minutes**

Marks: **308 marks**

Comments:

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q1.

$$f'(x) = \left(2x - \frac{3}{x}\right)^2 \text{ and } f(3) = 2$$

Find $f(x)$.

(Total 4 marks)

Q2.

$$p(x) = 30x^3 - 7x^2 - 7x + 2$$

(a) Prove that $(2x + 1)$ is a factor of $p(x)$

(2)

(b) Factorise $p(x)$ completely.

(3)

(c) Prove that there are no real solutions to the equation

$$\frac{30 \sec^2 x + 2 \cos x}{7} = \sec x + 1$$

(5)

(Total 10 marks)

Q3.

$$p(x) = x^3 - 5x^2 + 3x + a, \text{ where } a \text{ is a constant.}$$

Given that $x - 3$ is a factor of $p(x)$, find the value of a

Circle your answer. ©2025 Exam Papers Practice. All rights reserved.

−9

−3

3

9

(Total 1 mark)

Q4.

$$p(x) = 2x^3 + 7x^2 + 2x - 3$$

(a) Use the factor theorem to prove that $x + 3$ is a factor of $p(x)$

(2)

(b) Simplify the expression $\frac{2x^3 + 7x^2 + 2x - 3}{4x^2 - 1}, x \neq \pm \frac{1}{2}$

(4)

(Total 6 marks)

Q5.

The polynomial $p(x)$ is given by

$$p(x) = x^3 - 4x^2 - 3x + 18$$

- (a) (i) Use the Factor Theorem to show that $x - 3$ is a factor of $p(x)$. (2)
- (ii) Express $p(x)$ as a product of linear factors. (3)
- (b) Sketch the curve with equation $y = x^3 - 4x^2 - 3x + 18$, stating the values of x where the curve meets the x -axis. (3)

(Total 8 marks)

Q6.

The polynomial $f(x)$ is defined by $f(x) = 2x^3 + x^2 - 8x - 7$.

- (a) Find the remainder when $f(x)$ is divided by $(2x + 1)$. (2)
- (b) The polynomial $g(x)$ is defined by $g(x) = f(x) + d$, where d is a constant.
- (i) Given that $(2x + 1)$ is a factor of $g(x)$, show that $g(x) = 2x^3 + x^2 - 8x - 4$. (1)
- (ii) Given that $g(x)$ can be written as $g(x) = (2x + 1)(x^2 + a)$, where a is an integer, express $g(x)$ as a product of three linear factors. (1)
- (iii) Hence, or otherwise, show that $\frac{g(x)}{2x^3 - 3x^2 - 2x} = p + \frac{q}{x}$, where p and q are integers. (3)

(Total 7 marks)

Q7.

- (a) The polynomial $f(x)$ is given by $f(x) = x^3 - 4x + 15$.
- (i) Use the Factor Theorem to show that $x + 3$ is a factor of $f(x)$. (2)
- (ii) Express $f(x)$ in the form $(x + 3)(x^2 + px + q)$, where p and q are integers. (2)

(b) A curve has equation $y = x^4 - 8x^2 + 60x + 7$.

(i) Find $\frac{dy}{dx}$.

(3)

(ii) Show that the x -coordinates of any stationary points of the curve satisfy the equation

$$x^3 - 4x + 15 = 0$$

(1)

(iii) Use the results above to show that the only stationary point of the curve occurs when $x = -3$.

(2)

(iv) Find the value of $\frac{d^2y}{dx^2}$ when $x = -3$.

(3)

(v) Hence determine, with a reason, whether the curve has a maximum point or a minimum point when $x = -3$.

(1)

(Total 14 marks)

Q8.

The polynomial $f(x)$ is defined by $f(x) = 4x^3 - 11x - 3$.

(a) Use the Factor Theorem to show that $(2x + 3)$ is a factor of $f(x)$.

(2)

(b) Write $f(x)$ in the form $(2x + 3)(ax^2 + bx + c)$, where a , b and c are integers.

(2)

(c) (i) Show that the equation $2 \cos 2\theta \sin \theta + 9 \sin \theta + 3 = 0$ can be written as $4x^3 - 11x - 3 = 0$, where $x = \sin \theta$.

(3)

(ii) Hence find all solutions of the equation $2 \cos 2\theta \sin \theta + 9 \sin \theta + 3 = 0$ in the interval $0^\circ < \theta < 360^\circ$, giving your solutions to the nearest degree.

(4)

(Total 11 marks)

Q9.

(a) Use the Factor Theorem to show that $4x - 3$ is a factor of

$$16x^3 + 11x - 15$$

(2)

- (b) Given that $x = \cos \theta$, show that the equation

$$27 \cos \theta \cos 2\theta + 19 \sin \theta \sin 2\theta - 15 = 0$$

can be written in the form

$$16x^3 + 11x - 15 = 0$$

(4)

- (c) Hence show that the only solutions of the equation

$$27 \cos \theta \cos 2\theta + 19 \sin \theta \sin 2\theta - 15 = 0$$

are given by $\cos \theta = \frac{3}{4}$.

(4)

(Total 10 marks)

Q10.

The polynomial $p(x)$ is given by

$$p(x) = x^3 + 2x^2 - 5x - 6$$

- (a) (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$.

(2)

- (ii) Express $p(x)$ as the product of three linear factors.

(3)

- (b) Verify that $p(0) > p(1)$.

(2)

- (c) Sketch the curve with equation $y = x^3 + 2x^2 - 5x - 6$, indicating the values where the curve crosses the x -axis.

(3)

(Total 10 marks)

Q11.

- (a) (i) Express $\frac{5x-6}{x(x-3)}$ in the form $\frac{A}{x} + \frac{B}{x-3}$.

(2)

- (ii) Find $\int \frac{5x-6}{x(x-3)} dx$.

(2)

- (b) (i) Given that

$$4x^3 + 5x - 2 = (2x + 1)(2x^2 + px + q) + r$$

find the values of the constants p , q and r .

(4)

(ii) Find $\int \frac{4x^3 + 5x - 2}{2x + 1} dx$.

(3)

(Total 11 marks)

Q12.

- (a) (i) Sketch the curve with equation $y = x(x - 2)^2$.

(3)

- (ii) Show that the equation $x(x - 2)^2 = 3$ can be expressed as

$$x^3 - 4x^2 + 4x - 3 = 0$$

(1)

- (b) The polynomial $p(x)$ is given by $p(x) = x^3 - 4x^2 + 4x - 3$.

- (i) Use the Factor Theorem to show that $x - 3$ is a factor of $p(x)$.

(2)

- (ii) Express $p(x)$ in the form $(x - 3)(x^2 + bx + c)$, where b and c are integers.

(2)

- (c) Hence show that the equation $x(x - 2)^2 = 3$ has only one real root and state the value of this root.

(3)

(Total 11 marks)

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q13.

- (a) The polynomial $f(x)$ is defined by $f(x) = 9x^3 + 18x^2 - x - 2$.

- (i) Use the Factor Theorem to show that $3x + 1$ is a factor of $f(x)$.

(2)

- (ii) Express $f(x)$ as a product of three linear factors.

(3)

(iii) Simplify $\frac{9x^3 + 21x^2 + 6x}{f(x)}$.

(3)

- (b) When the polynomial $9x^3 + px^2 - x - 2$ is divided by $3x - 2$, the remainder is -4 .

Find the value of the constant p .

(2)

(Total 10 marks)

Q14.

The polynomial $p(x)$ is given by $p(x) = x^3 - 2x^2 + 3$.

- (a) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$.

(2)

- (b) (i) Express $p(x) = x^3 - 2x^2 + 3$ in the form $(x + 1)(x^2 + bx + c)$, where b and c are integers.

(2)

- (ii) Hence show that the equation $p(x) = 0$ has exactly one real root.

(2)

(Total 6 marks)

Q15.

The polynomial $f(x)$ is defined by $f(x) = 4x^3 - 13x + 6$.

- (a) Find $f(-2)$.

(1)

- (b) Use the Factor Theorem to show that $2x - 3$ is a factor of $f(x)$.

(2)

- (c) Simplify $\frac{2x^2 + x - 6}{f(x)}$.

(4)

(Total 7 marks)

Q16.

- (a) Expand $(5 + h)^3$.

(1)

- (b) A curve has equation $y = x^3 - x^2$.

- (i) Find the gradient of the line passing through the point $(5, 100)$ and the point on the curve for which $x = 5 + h$. Give your answer in the form

$$p + qh + rh^2$$

where p , q and r are integers.

(4)

- (ii) Show how the answer to part (b)(i) can be used to find the gradient of the

curve at the point (5, 100). State the value of this gradient.

(2)

(Total 7 marks)

Q17.

The polynomial $p(x)$ is given by $p(x) = x^3 - 13x - 12$.

(a) Use the Factor Theorem to show that $x + 3$ is a factor of $p(x)$.

(2)

(b) Express $p(x)$ as the product of three linear factors.

(3)

(Total 5 marks)

Q18.

The polynomial $f(x)$ is defined by $f(x) = 15x^3 + 19x^2 - 4$.

(a) (i) Find $f(-1)$.

(1)

(ii) Show that $(5x - 2)$ is a factor of $f(x)$.

(2)

(b) Simplify

$$\frac{15x^2 - 6x}{f(x)}$$

giving your answer in a fully factorised form.

(5)

(Total 8 marks)

©2025 Exam Papers Practice. All rights reserved.

Q19.

The expression $\frac{10x^2 + 8}{(x+1)(5x-1)}$ can be written in the form $2 + \frac{A}{x+1} + \frac{B}{5x-1}$, where A and B are constants.

(a) Find the values of A and B .

(4)

(b) Hence find $\int \frac{10x^2 + 8}{(x+1)(5x-1)} dx$.

(4)

(Total 8 marks)

Q20.

The polynomial $p(x)$ is given by

$$p(x) = x^3 + 7x^2 + 7x - 15$$

- (a) (i) Use the Factor Theorem to show that $x + 3$ is a factor of $p(x)$. (2)
- (ii) Express $p(x)$ as the product of three linear factors. (3)
- (b) (i) Verify that $p(-1) < p(0)$. (1)
- (ii) Sketch the curve with equation $y = x^3 + 7x^2 + 7x - 15$, indicating the values where the curve crosses the coordinate axes. (4)

(Total 10 marks)

Q21.

The polynomial $g(x)$ is defined by $g(x) = 8x^3 + 6x^2 - 14x + d$.

- (a) Given that $(4x - 1)$ is a factor of $g(x)$, find the value of the constant d . (2)
- (b) Given that $g(x) = (4x - 1)(ax^2 + bx + c)$, find the values of the integers a , b and c . (3)

(Total 5 marks)

Q22.

- (a) (i) Express $\frac{7x-3}{(x+1)(3x-2)}$ in the form $\frac{A}{x+1} + \frac{B}{3x-2}$. (3)

- (ii) Hence find $\int \frac{7x-3}{(x+1)(3x-2)} dx$. (2)

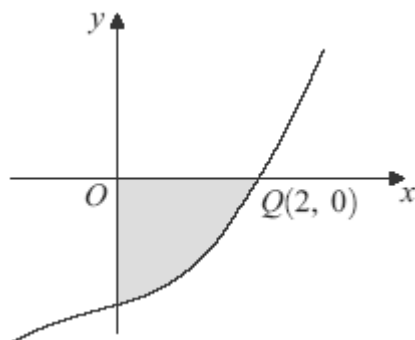
- (b) Express $\frac{6x^2 + x + 2}{2x^2 - x + 1}$ in the form $P + \frac{Qx + R}{2x^2 - x + 1}$. (3)

(Total 8 marks)

Q23.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 + x - 10$.

- (i) Use the Factor Theorem to show that $x - 2$ is a factor of $p(x)$. (2)
- (ii) Express $p(x)$ in the form $(x - 2)(x^2 + ax + b)$, where a and b are constants. (2)
- (b) The curve C with equation $y = x^3 + x - 10$, sketched below, crosses the x -axis at the point $Q(2, 0)$.



- (i) Find the gradient of the curve C at the point Q . (4)
- (ii) Hence find an equation of the tangent to the curve C at the point Q . (2)
- (iii) Find $\int (x^3 + x - 10) dx$. (3)
- (iv) Hence find the area of the shaded region bounded by the curve C and the coordinate axes. (2)

(Total 15 marks)

©2025 Exam Papers Practice. All rights reserved.

Q24.

The polynomial $f(x)$ is defined by $f(x) = 4x^3 - 7x - 3$.

- (a) Find $f(-1)$. (1)
- (b) Use the Factor Theorem to show that $2x + 1$ is a factor of $f(x)$. (2)
- (c) Simplify the algebraic fraction $\frac{4x^3 - 7x - 3}{2x^2 + 3x + 1}$. (3)

(Total 6 marks)

Q25.

- (a) (i) Express $\frac{2x+7}{x+2}$ in the form $A + \frac{B}{x+2}$, where A and B are integers. (2)

- (ii) Hence find $\int \frac{2x+7}{x+2} dx$. (2)

- (b) (i) Express $\frac{28+4x^2}{(1+3x)(5-x)^2}$ in the form $\frac{P}{1+3x} + \frac{Q}{5-x} + \frac{R}{(5-x)^2}$, where P , Q and R are constants. (5)

- (ii) Hence find $\int \frac{28+4x^2}{(1+3x)(5-x)^2} dx$. (4)

(Total 13 marks)

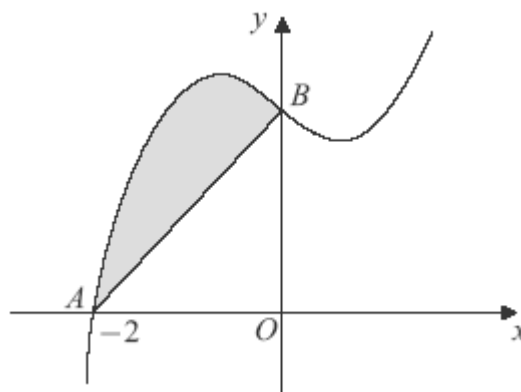
Q26.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - x + 6$.
 (i) Use the Factor Theorem to show that $x + 2$ is a factor of $p(x)$. (2)

- (ii) Express $p(x) = x^3 - x + 6$ in the form $(x + 2)(x^2 + bx + c)$, where b and c are integers. (2)

- (iii) The equation $p(x) = 0$ has one root equal to -2 . Show that the equation has no other real roots. (2)

- (b) The curve with equation $y = x^3 - x + 6$ is sketched below.



The curve cuts the x -axis at the point $A(-2, 0)$ and the y -axis at the point B .

- (i) State the y -coordinate of the point B .

(1)

- (ii) Find $\int_{-2}^0 (x^3 - x + 6) dx$.

(5)

- (iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - x + 6$ and the line AB .

(3)

(Total 15 marks)

Q27.

Express $\frac{3x^3 + 8x^2 - 3x - 5}{3x - 1}$ in the form $ax^2 + bx + \frac{c}{3x - 1}$, where a , b and c are integers.

(Total 3 marks)

Q28.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - 7x - 6$.

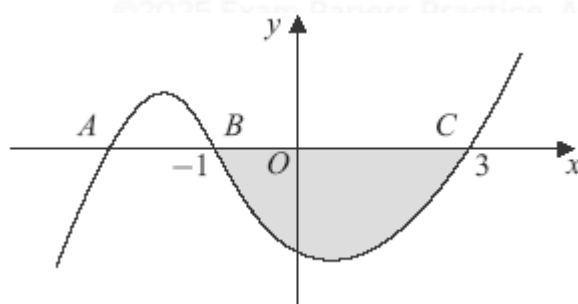
- (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$.

(2)

- (ii) Express $p(x) = x^3 - 7x - 6$ as the product of three linear factors.

(3)

- (b) The curve with equation $y = x^3 - 7x - 6$ is sketched below.



The curve cuts the x -axis at the point A and the points $B(-1, 0)$ and $C(3, 0)$.

- (i) State the coordinates of the point A .

(1)

- (ii) Find $\int_{-1}^3 (x^3 - 7x - 6) dx$.

(5)

- (iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - 7x - 6$ and the x -axis between B and C .

(1)

- (iv) Find the gradient of the curve $y = x^3 - 7x - 6$ at the point B .

(3)

- (v) Hence find an equation of the normal to the curve at the point B .

(3)

(Total 18 marks)

Q29.

- (a) The polynomial $f(x)$ is defined by $f(x) = 2x^3 + 3x^2 - 18x + 8$.

- (i) Use the Factor Theorem to show that $(2x - 1)$ is a factor of $f(x)$.

(2)

- (ii) Write $f(x)$ in the form $(2x - 1)(x^2 + px + q)$, where p and q are integers.

(2)

- (iii) Simplify the algebraic fraction $\frac{4x^2 + 16x}{2x^3 + 3x^2 - 18x + 8}$.

(2)

- (b) Express the algebraic fraction $\frac{2x^2}{(x+5)(x-3)}$ in the form $A + \frac{B+Cx}{(x+5)(x-3)}$, where A , B and C are integers.

(4)

(Total 10 marks)

©2025 Exam Papers Practice. All rights reserved.

Q30.

- (a) (i) Use the Factor Theorem to show that $x + 2$ is a factor of $p(x)$.

(2)

- (ii) Express $p(x)$ as the product of linear factors.

(3)

- (b) (i) The curve with equation $y = x^3 + x^2 - 8x - 12$ passes through the point $(0, k)$. State the value of k .

(1)

- (ii) Sketch the graph of $y = x^3 + x^2 - 8x - 12$, indicating the values of x where the curve touches or crosses the x -axis.

(3)

(Total 9 marks)

Q31.

The polynomial $f(x)$ is defined by $f(x) = 27x^3 - 9x + 2$.

(a) Show that $f\left(-\frac{2}{3}\right) = 0$. (1)

(b) Express $f(x)$ as a product of three linear factors. (4)

(c) Simplify

$$\frac{27x^3 - 9x + 2}{9x^2 + 3x - 2}$$

(2)

(Total 7 marks)

Q32.

The polynomial $p(x)$ is given by

$$p(x) = x^3 - 4x^2 - 7x + k$$

where k is a constant.

(a) (i) Given that $x + 2$ is a factor of $p(x)$, show that $k = 10$. (2)

(ii) Express $p(x)$ as the product of three linear factors. (3)

(b) Sketch the curve with equation $y = x^3 - 4x^2 - 7x + 10$, indicating the values where the curve crosses the x -axis and the y -axis. (You are **not** required to find the coordinates of the stationary points.)

(4)

(Total 9 marks)

Q33.

The polynomial $f(x)$ is defined by $f(x) = 2x^3 - 7x^2 + 13$.

(a) The polynomial $g(x)$ is defined by $g(x) = 2x^3 - 7x^2 + 13 + d$, where d is a constant. Given that $(2x - 3)$ is a factor of $g(x)$, show that $d = -4$.

(2)

(b) Express $g(x)$ in the form $(2x - 3)(x^2 + ax + b)$.

(2)
(Total 4 marks)

Q34.

- (a) (i) Express $\frac{3x-5}{x-3}$ in the form $A + \frac{B}{x-3}$, where A and B are integers. (2)

- (ii) Hence find $\int \frac{3x-5}{x-3} dx$. (2)

- (b) (i) Express $\frac{6x-5}{4x^2-25}$ in the form $\frac{P}{2x+5} + \frac{Q}{2x-5}$, where P and Q are integers. (3)

- (ii) Hence find $\int \frac{6x-5}{4x^2-25} dx$. (3)
- (Total 10 marks)

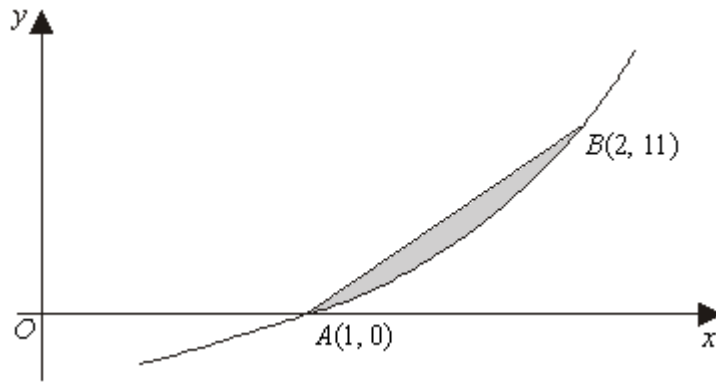
Q35.

- (a) The polynomial $f(x)$ is given by $f(x) = x^3 + 4x - 5$.
- (i) Use the Factor Theorem to show that $x - 1$ is a factor of $f(x)$ (2)

- (ii) Express $f(x)$ in the form $(x - 1)(x^2 + px + q)$, where p and q are integers. (2)

- (iii) Hence show that the equation $f(x) = 0$ has exactly one real root and state its value. (3)

- (b) The curve with equation $y = x^3 + 4x - 5$ is sketched below.



The curve cuts the x -axis at the point $A(1, 0)$ and the point $B(2, 11)$ lies on the curve.

(i) Find $\int (x^3 + 4x - 5) \, dx$.

(3)

(ii) Hence find the area of the shaded region bounded by the curve and the line AB .

(4)

(Total 14 marks)



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.